

Erdős Problem #811

EP-811

L1 - TRACTABLE

PARTIALLY SOLVED

Graph Theory

ORIGINAL PROBLEM STATEMENT

Suppose $n \equiv 1 \pmod{m}$. We say that an edge-colouring of K_n using m colours is balanced if every vertex sees exactly $\lfloor n/m \rfloor$ many edges of each colour.

For which graphs G is it true that, if $m = e(G)$, for all large $n \equiv 1 \pmod{m}$, every balanced edge-colouring of K_n with m colours contains a rainbow copy of G ? (That is, a subgraph isomorphic to G where each edge receives a different colour.)

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